

### Density processing

**1** Raw active density  $\gamma$

**2** Active-mask Helmholtz filter  $\tilde{\gamma}$

**3** Projection and passive bounds  $\bar{\gamma}$

**4** Material fields  $\alpha(\bar{\gamma})$   
SA and wall penalties

### Forward turbulent Picard loop

**8** Picard residual accepted?

**7** SA transport  $\tilde{\nu}$  and  $\nu_t$

**6** Flow solve IPCS/SNES  $\mathbf{u}, p$

**5** Reciprocal wall distance  $G, y(G)$

repeat until accepted

accepted

**9** Final frozen state solve  $\mathbf{s} = (\mathbf{u}, p)$

**10** Frozen flow adjoint  $\mathbf{s}^*$

**11** Sensitivities  $dJ, dV$  with respect to  $\bar{\gamma}$

**12** Projection slope and transpose filter

**13** MMA update active raw DOFs

### Frozen sensitivity and optimizer update

next MMA iteration